# Fine-tuning AlphaGenome on single-cell pseudobulk multiome data

## 1 Summary

One AlphaGenome [1] model was fine-tuned jointly on five single-cell multiome studies (ten species-specific sources, 20 output heads). It starts from the AlphaGenome FOLD_1 checkpoint and is evaluated on genome regions that this checkpoint never saw as pretraining targets. The most effective changes were to the training targets: RNA rebuilt as gene-level pseudobulk counts per million (CPM) from raw counts, ATAC rebuilt from raw fragments where released tracks were unsuitable, and removal of near-empty, unsupported or artifactual signal. On held-out regions, frozen-backbone heads reach mean correlation $R = 0.455$, and adding low-rank adapters raises it to 0.500–0.507 (ATAC 0.66–0.67, RNA 0.33–0.35). LoRA and LoRA with convolutional adapters (LoCon) perform similarly. On fine-mapped brain eQTLs, gene-span variant effects give AUROC up to 0.57 (SingleBrain) and 0.64 (XQTL). The frozen-backbone model is best for eQTLs, adapter training lowers eQTL discrimination, and discrimination is concentrated within 10 kb of the TSS (Figure 1).

An earlier version of this report fine-tuned from the `all_folds` checkpoint and held out chromosomes 8 and 9, which that checkpoint was pretrained on. Its held-out results (mean test $R$ 0.627) are superseded.

## 2 Data

Each study provides pseudobulk ATAC and RNA profiles for annotated cell groups (Table 1) [3, 4, 5, 6, 7]. Each source keeps its own genome assembly. Macaque and marmoset sequences use AlphaGenome's human organism embedding.

Table 1: Training data. Group counts are ATAC/RNA cell groups; Liu and Johansen RNA heads have one channel per strand. ATAC source is the released track or a reconstruction from raw fragments.

| Study | Tissue | Species | Groups | ATAC source |
|---|---|---|---|---|
| Mannens (HDA) [3] | first-trimester brain | human | 134/134 | released |
| Liu (HDMA) [4] | fetal, 12 organs | human | 186/186 | fragments |
| Johansen 2025 [5] | basal ganglia | human, macaque, marmoset | 47/47 | fragments |
| Zemke 2023 [6] | motor cortex | human, macaque, marmoset, mouse | 20/19 | released |
| Zemke 2024 [7] | hippocampus (aging) | human | 18/18 | fragments |

**Held-out regions.** AlphaGenome's FOLD_1 checkpoint [1] was pretrained without two of the eight Borzoi genome folds [2], one used here for validation and one for test. Input windows of 131,072 bp are assigned to training, validation or test by these folds, directly for human and mouse and through liftover of 4 kb bins to hg38 for macaque and marmoset, so that sequence orthologous

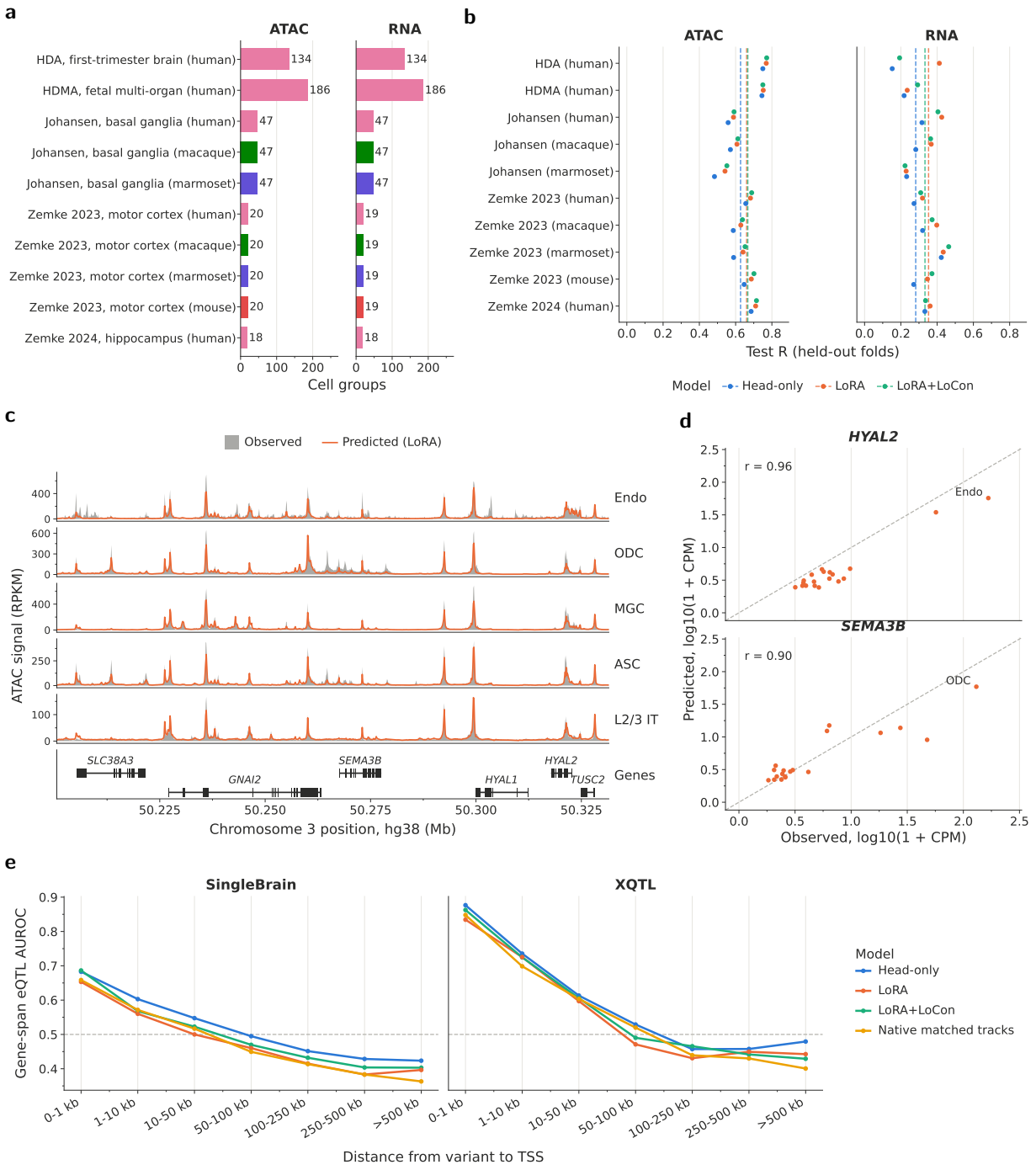



Figure 1: Training data and performance of the FOLD_1 fine-tuned models. (a) Pseudobulk cell groups per source for ATAC and RNA, colored by species. (b) Test $R$ per source on FOLD_1 held-out regions with the Zemke 2023 ATAC blacklist applied. Dashed lines are means across sources. (c) Observed and predicted (LoRA) ATAC for five of 20 cell groups in a held-out 131 kb window of the Zemke 2023 human motor cortex source. Predictions are the 1 bp output averaged over 128 bp bins, in the units of the observed track. The window was picked from four held-out windows containing cell-type-specific genes, and its ATAC $R$ (0.83) is above the source mean (0.68). (d) Observed against predicted gene-level RNA across cell groups for two genes in the window in (c). Predicted CPM is the exon-weighted sum of 128 bp predictions, and $r$ is the Pearson correlation across cell groups. The gray line is $y = x$. (e) Gene-span AUROC for fine-mapped brain eQTLs by distance between variant and TSS. Native matched tracks are AlphaGenome's own tracks, selected on the same variants, so their values are optimistic. The dashed line is chance.

to held-out human regions is also held out. Windows that overlap more than one split are dropped (0.4–4.4% per source), and validation and test windows are subsampled to about 1,000 per source.

## 3 Training strategies

Performance is measured by signed double-centered Pearson correlation $R$. For target and prediction matrices $Y, \hat{Y} \in \mathbb{R}^{N \times C}$, with $N$ genomic elements (genes for RNA, 128 bp bins for ATAC) and $C$ cell groups, row and column means are removed before computing the correlation, so $R$ measures relative structure across elements and cell groups rather than assay scale. The comparisons below come from matched pairs of earlier runs that differed in one setting; those runs used the chromosome split, so their absolute values are inflated but the differences between arms are informative.

**RNA targets and loss.** Released coordinate RNA tracks were replaced with gene-level targets. Raw counts are summed over the cells of each group before CPM normalization; summing per-cell normalized expression instead gives targets whose double-centered agreement with the correct ones is only 0.46–0.54 (Johansen). The model's RNA output $\hat{x} \in \mathbb{R}^{B \times C}$ over $B$ 128 bp bins is read out per gene $g$ as

$$\hat{y}_{g,c} = \sum_b w_{g,b}\, \hat{x}_{b,c}, \tag{1}$$

where $w_{g,b} \in [0, 1]$ is the fraction of bin $b$ covered by exons of $g$, and the loss is the squared error between $\log(1 + \hat{y}_{g,c})$ and $\log(1 + y_{g,c})$. In a matched frozen-backbone comparison on Zemke 2023 that changed only this target, test RNA $R$ rose from 0.152 to 0.421 with ATAC unchanged. The gene-level target also avoids a ribosomal-repeat artifact in the released marmoset RNA tracks. Adding a per-bin coverage loss did not help (HDA, mean validation $R$ 0.604 with versus 0.598 without), so it was dropped.

**ATAC targets.** Where released tracks were significance scores or poorly matched coverage, ATAC was rebuilt from raw fragments as full-depth fragment coverage per million fragments, without downsampling. For Johansen human, test $R$ was 0.210 with released tracks and 0.458 with the rebuilt target under the same training setup.

**Quality control.** One Johansen group with 33,424 fragments (median $1.1 \times 10^8$) contributed 48% of all squared ATAC signal and was removed. Groups without supporting cell assignments are masked rather than given zero targets: four Zemke 2024 subtype channels and the Zemke 2023 L5 IT RNA group. The released Zemke 2023 ATAC tracks contain bins with signal of 3,000 to 60,000 RPKM in every cell group, mostly poly-G/C runs, against genome-wide 99.99th percentiles of 97 to 169. In the macaque and marmoset held-out sets a single such bin or window carried 51–89% of the double-centered variance, so validation and test $R$ for these heads measured one or two loci. Following the idea of the ENCODE blacklist [10], a blacklist of bins whose minimum across cell groups exceeds ten times the 99.99th percentile, and of windows whose summed signal exceeds ten times the median window, flags 9 to 19 regions per species. Excluding them removes one held-out window per affected split and leaves no held-out window with more than 4% of the variance. The blacklist is applied at evaluation; the training windows still contain these regions.

**Loss.** A double-centered correlation term, $1 - R$ computed within each batch, was added to the loss of every RNA head (weight 1, HDA 0.1) and of the Zemke 2023 ATAC heads (weight 3). In a

matched continuation it raised validation $R$ from 0.568 to 0.583, mainly through RNA (0.461 to 0.491).

**Model and optimization.** The pretrained backbone is shared by all heads. Each source has its own ATAC and RNA heads, because target distributions differ between species of the same study; with equal sampling of sources this raised LoRA test $R$ from 0.584 to 0.605 relative to one pooled head per study and modality, a comparison that changes both factors together. Randomly initialized heads are first trained on a frozen backbone; the best warmup epoch then starts adapter training. Adapters are LoRA [8] (rank 16) on attention and pair projections, optionally with LoCon [9] (rank 4) on downsampling blocks 2–5, at learning rate $3 \times 10^{-4}$; learning rate $10^{-3}$ was unstable for LoCon. Initializing heads from pretrained neural accessibility tracks improved early warmup validation $R$ (0.507 against 0.488 at epoch 4) but was not used in the final runs.

**Not helpful.** Higher adapter rank (LoRA 32, LoCon 8), uniform RNA loss upweighting, target value transforms, refitting heads after adapter training, and gene-balanced batch ordering did not improve held-out $R$.

## 4 Held-out results

The frozen-backbone warmup selected epoch 5 by validation $R$. Both adapter arms were trained from it to epoch 18 and selected epoch 18 (LoRA) and epoch 15 (LoRA+LoCon). Table 2 lists test $R$ per source with the Zemke 2023 ATAC blacklist applied. Adapters raise mean test $R$ from 0.455 to 0.507 (LoRA) and 0.500 (LoRA+LoCon). LoRA+LoCon is slightly better for ATAC (0.667 against 0.661) and worse for RNA (0.333 against 0.353), so the two adapter strategies are effectively tied. RNA remains the weaker modality and varies between sources, from about 0.2 (Johansen marmoset, HDA with LoCon, Liu) to 0.46 (Zemke 2023 marmoset). Split-half reliability gives ATAC correlation ceilings of 0.82–0.93 for Johansen and 0.90 for Zemke 2024, and RNA ceilings of 0.93 or higher where measured, so RNA is far below its noise limit.

Table 2: Test $R$ on FOLD_1 held-out regions with the Zemke 2023 ATAC blacklist applied.

| | | Head-only | | LoRA | | LoRA+LoCon | |
|---|---|---|---|---|---|---|---|
| Study | Species | ATAC | RNA | ATAC | RNA | ATAC | RNA |
| HDA | human | 0.750 | 0.152 | 0.769 | 0.412 | 0.771 | 0.193 |
| Liu | human | 0.745 | 0.218 | 0.752 | 0.235 | 0.750 | 0.293 |
| Johansen 2025 | human | 0.559 | 0.316 | 0.588 | 0.426 | 0.591 | 0.405 |
| | macaque | 0.571 | 0.282 | 0.607 | 0.367 | 0.613 | 0.364 |
| | marmoset | 0.483 | 0.232 | 0.541 | 0.229 | 0.552 | 0.223 |
| Zemke 2023 | human | 0.656 | 0.273 | 0.682 | 0.319 | 0.689 | 0.310 |
| | macaque | 0.587 | 0.320 | 0.629 | 0.398 | 0.638 | 0.372 |
| | marmoset | 0.589 | 0.423 | 0.641 | 0.434 | 0.652 | 0.464 |
| | mouse | 0.647 | 0.271 | 0.687 | 0.347 | 0.701 | 0.372 |
| Zemke 2024 | human | 0.685 | 0.333 | 0.710 | 0.362 | 0.715 | 0.336 |
| Mean | | 0.627 | 0.282 | 0.661 | 0.353 | 0.667 | 0.333 |

## 5 Fine-mapped eQTL benchmark

Positives are variants with SuSiE [11] posterior inclusion probability (PIP) above 0.75 and negatives have PIP below 0.01, within 512 kb of the gene's TSS. SingleBrain [13], a single-nucleus eQTL meta-analysis of 983 individuals whose summary statistics are labeled as mmQTL [12] analyses, contributes 35 fine-mapping panels in seven brain cell classes (54,193 variants). The ROSMAP single-nucleus xQTL resource [14] contributes six cell-type panels (10,452 variants). Each variant is scored with the human Zemke 2023 RNA head on a 1 Mb window centered at the TSS. The gene-span effect is the absolute alternate-minus-reference difference in mean prediction over the gene's bins, averaged over the tracks matched to the panel's cell type. The TSS-bin summary performs worse and is not shown.

Table 3 compares the three models with one selected pretrained AlphaGenome track per cell type. The frozen-backbone model is best on both datasets, and adapter training lowers pooled AUROC by 0.01–0.06. The native tracks were selected on the same variants, so their values are optimistic. Discrimination is concentrated near the TSS: AUROC is 0.65–0.69 (SingleBrain) and 0.83–0.88 (XQTL) within 1 kb and falls below 0.5 for all models beyond about 50 kb (SingleBrain) and 100 kb (XQTL) (Figure 2).

Table 3: Gene-span eQTL discrimination. AUROC and AP are pooled over panels; "within" is the mean of per-cell-class AUROC.

| | SingleBrain | | | XQTL | | |
|---|---|---|---|---|---|---|
| Model | AUROC | AP | Within | AUROC | AP | Within |
| Head-only | 0.566 | 0.472 | 0.559 | 0.644 | 0.636 | 0.648 |
| LoRA | 0.511 | 0.407 | 0.501 | 0.614 | 0.591 | 0.597 |
| LoRA+LoCon | 0.532 | 0.433 | 0.527 | 0.631 | 0.608 | 0.624 |
| Native matched tracks | 0.524 | 0.439 | 0.535 | 0.627 | 0.626 | 0.649 |

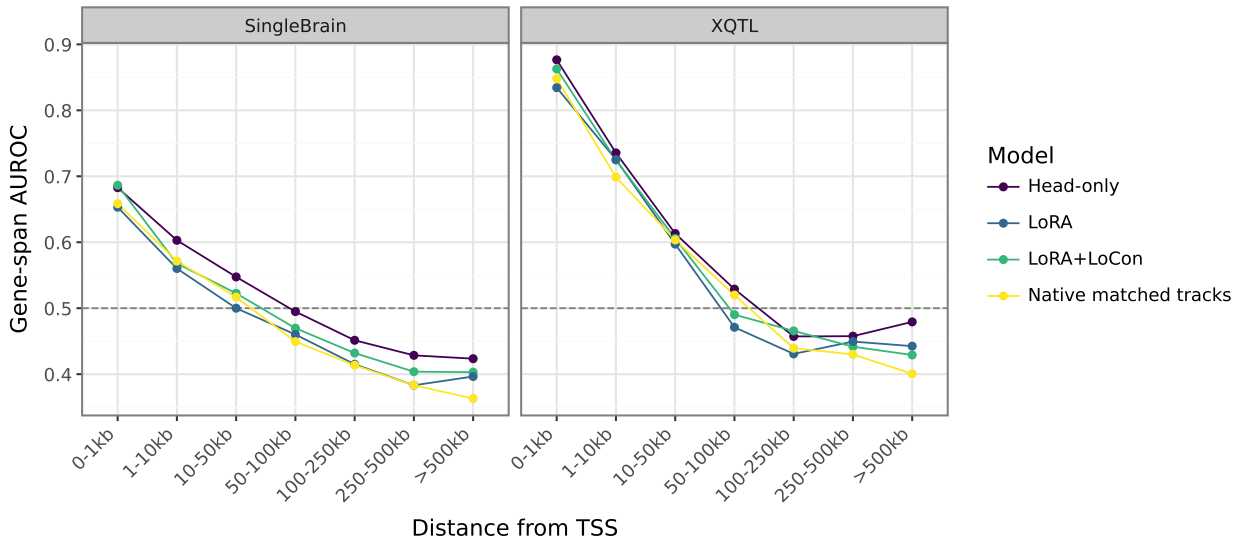



Figure 2: Gene-span AUROC by distance between variant and TSS. The dashed line is chance.

The trained tracks carry a small amount of cell-type-specific signal. Scoring each panel with

every cell class's tracks, the matched class exceeds the unmatched ones by 0.012–0.028 AUROC on SingleBrain and 0.039–0.044 on XQTL, which is significant under permutation of the class assignment for every model ($p \leq 0.006$). The matched composite ranks first for microglia panels in every model and dataset and for most astrocyte and oligodendrocyte panels, whereas excitatory and inhibitory neuron panels show little specificity. Predicted effects are largely shared across cell types, with median Spearman correlation 0.61–0.72 between tracks.

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